

AI for Medicine – Project Report

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1. Project Title

3D Volumetric Detection of Alzheimer’s Disease from Structural MRI: A Comparative Study on the OASIS-1 Dataset

2. Problem Statement

Alzheimer’s Disease (AD) is the primary cause of dementia globally, affecting over 55 million individuals. Early and accurate diagnosis is crucial because clinical interventions and disease-modifying therapies are significantly more effective during the earliest symptomatic stages. While structural T1-weighted MRI is a clinical standard used to identify structural brain atrophy (particularly within the hippocampus and medial temporal lobe) visual radiological evaluation remains dependent on the observer's experience.

Developing an automated, image-based machine learning decision-support system can establish clinical consistency and serve as an accessible screening tool in non-specialist clinical environments. However, training robust deep learning models on real-world clinical cohorts presents major technical challenges: available public datasets are typically limited in sample size, imbalanced, and socio-demographically heterogeneous. Under these constrained conditions, the rigorous configuration of the validation pipeline, rather than the raw complexity of the architecture, determines whether the experimental results reflect true clinical generalizability or statistical artifacts like data leakage.

3. Objective of the Study

The primary objective of this study is to develop a reliable machine learning pipeline to classify subjects from the cross-sectional OASIS-1 cohort into one of two diagnostic categories: Cognitively Healthy (CN) or Alzheimer’s Disease (AD), utilizing 3D T1-weighted structural MRI volumes.

To establish a rigorous benchmarks framework under a shared, identical validation protocol, we conduct a comprehensive comparative analysis across five distinct methodological approaches:

- A 3D Convolutional Neural Network trained from scratch (ResNet-18-3D).
- A deeper 3D CNN architecture trained from scratch (DenseNet-121-3D).
- A 3D CNN initialized via domain-specific medical pretraining (MedicalNet ResNet-18-PT).
- A multi-modal architecture fusing deep pretrained visual features with non-imaging clinical demographics (Age and Sex).
- A classical baseline (Random Forest) trained on tabular, FreeSurfer-derived volumetric features and demographics.

A secondary objective is to implement a clinically informed decision framework by comparing different operating thresholds and the Youden-optimal threshold, evaluating model suitability based on statistical stability and clinical utility rather than relying solely on arbitrary point performance estimates.

4. Dataset Description

- **Source:** Open Access Series of Imaging Studies (OASIS-1) Cross-Sectional cohort, publicly released for non-commercial scientific research under the official OASIS Data Use Agreement.
- **Composition and Cohort Selection:** The original release contains 436 cross-sectional records from 416 subjects. To construct a clinically valid and un-biased experimental cohort, strict inclusion criteria were enforced. We retained subjects with:
 1. A valid, non-missing Clinical Dementia Rating (CDR) to serve as the definitive target label.
 2. An age criterion of 58 years or older to eliminate young-control distribution bias and isolate the population naturally at risk for AD.
 3. A fully completed and error-free FreeSurfer volumetric processing pipeline.This rigorous filtering yielded a final structured cohort of 196 unique subjects (Healthy = 103, AD = 93).
- **Data Modality and Formats:** For the volumetric deep learning models, we used structural T1-weighted MRI scans processed via FreeSurfer. These volumes were standardized, brain-extracted, and aligned to the international Talairach 88 stereotaxic space with a uniform 1 mm isotropic resolution. For the tabular pipelines, the extracted features consisted of Age, Sex, Education (Educ), Socioeconomic Status (SES), Mini-Mental State Examination (MMSE), Estimated Total Intracranial Volume (eTIV), Normalized Whole-Brain Volume (nWBV), and Atlas Scaling Factor (ASF).

- **Available Labels:** Binary classification targets were derived directly from the clinical CDR scores where $\text{Label} = (\text{CDR} > 0)$. Specifically, a CDR of 0 denotes Cognitively Healthy, whereas a $\text{CDR} > 0$ (ranging from 0.5 to 2) denotes the presence of cognitive impairment/AD.
- **Dataset Challenges and Biases:** The primary constraint is the modest sample size ($n = 196$), which heavily restricts parameter optimization. A distinct demographic sex bias is present within the OASIS-1 data distribution, skewing toward female subjects (130 females vs. 66 males). Lastly, using the clinical CDR score as our absolute ground truth means our dataset's labels are affected by natural human inconsistency in medical assessments.

5. Data Preprocessing

The data preprocessing pipeline was developed deterministically and executed completely end-to-end prior to model partitioning, ensuring that operations were independent for each subject volume to eliminate cross-subject data contamination.

For each individual 3D structural MRI volume, the workflow proceeded as follows:

1. **Volumetric Ingestion:** The raw FreeSurfer .mgz volumes were dynamically loaded via MONAI's integrated LoadImaged utility backed by a NibabelReader core.
 2. **Spatial Standardization:** A channel-first tensor orientation was strictly enforced across all components.
 3. **Background Optimization:** Uninformative background air voxels were structurally discarded by executing a localized foreground crop operation (CropForegroundd) using a strict spatial padding margin of 5 voxels.
 4. **Isotropic Resampling:** The cropped volumes were spatially resized using trilinear interpolation to a standardized, uniform isotropic 3D shape of $128 \times 128 \times 128$ voxels, establishing a consistent input dimension for all evaluated deep learning networks.
 5. **Intensity Normalization:** Voxel intensity ranges were scaled per-image to a bounded floating-point interval of $[0, 1]$ via ScaleIntensityd. The fully preprocessed tensors were cached directly into memory to eliminate I/O runtime overhead and ensure execution consistency across architectures.
- **Train-Time Augmentation:** To mitigate overfitting within the training partitions, localized stochastic transformations were applied online during the training phase only (ensuring no leakage to validation/test sets):
 - Random Affine Transformations: Rotation limits of plus/minus 0.1 rad, translation bounds of plus/minus 3 voxels, and scaling factors of plus/minus 5% (execution probability $p = 0.5$).
 - Asymmetric Flips: Random left-right flipping along the sagittal axis ($p = 0.5$).
 - Noise Injection: Random additive Gaussian noise ($\text{sigma} = 0.005$, $p = 0.2$).
 - **Test-Time Controls:** Validation and testing loaders excluded all forms of stochastic data augmentation. During inference, Test-Time Augmentation (TTA) was introduced by computing the final probability output as the arithmetic mean of the predictions from the original volume and its horizontally flipped counterpart. For the Random Forest baseline,

tabular features were imputed using a fold-isolated median strategy (SimpleImputer) and standardized via StandardScaler.

6. Avoiding Data Leakage

Data leakage is a critical issue in medical ML; thus, multiple structural guardrails were implemented to guarantee evaluation integrity:

- **3D Volumetric Patient Isolation:** We utilized a holistic 3D approach rather than 2D slice-based methods. While 2D approaches treat cross-sectional slices as independent vectors (systematically leaking highly correlated intra-subject morphology across validation splits) our 3D framework processes each patient as a single independent sample.
- **Isolated Partitioning & Tuning:** A 5-fold cross-validation split (StratifiedKFold) was instantiated globally. Inside each loop, a 25% stratified partition was carved exclusively from the training pool to monitor early stopping and tune hyperparameters. This ensured the test fold remained completely unseen during model design.
- **Fold-Isolated Normalization:** Tabular imputation (SimpleImputer) and scaling (StandardScaler) mechanisms were fit strictly on the designated training rows of each active fold before transforming the unseen test slice.
- **Thresholding Guardrails:** To prevent individual folds from "peeking" to adjust localized classification bounds, the Youden-optimal decision threshold was calculated post-hoc based on the fully pooled, completely out-of-fold (OOF) prediction array ($n = 196$).

7. Machine Learning Pipeline

Validation and Ensembling Protocol

The validation architecture follows a 5-fold stratified cross-validation strategy, stratified by target CDR labels, ensuring identical fold compositions across all models. Within each isolated fold, a 3-seed ensemble mechanism was implemented utilizing distinct initializations (seeds 42, 7, and 21). The final inference probability for any given fold was generated by computing the arithmetic mean of the individual seed network outputs before performance scoring. This structural design required training a total of 75 distinct machine learning models (60 deep neural networks and 15 Random Forest classifiers), providing substantial resistance against stochastic parameter initialization variance. Class imbalance was handled by applying localized loss penalization weights computed dynamically from the active training fold distributions for CNNs, and setting `class_weight='balanced'` for the Random Forest classifier.

Model Specifications

The structural configurations, backbones, and input specifications across the five evaluated pipelines are outlined below:

- **ResNet-18-3D (Scratch):** MONAI 3D ResNet-18 implementation with Spatial Dropout ($p = 0.5$). Initialized randomly. Inputs: Preprocessed 3D MRI Volume ($1 \times 128 \times 128 \times 128$).

- **DenseNet-121-3D:** MONAI 3D DenseNet-121 implementation with Feature Dropout ($p = 0.2$). Initialized randomly. Inputs: Preprocessed 3D MRI Volume ($1 \times 128 \times 128 \times 128$).
- **ResNet-18-PT:** MONAI 3D ResNet-18 initialized with specialized MedicalNet weights (102 of 125 layers loaded). Inputs: Preprocessed 3D MRI Volume ($1 \times 128 \times 128 \times 128$).
- **Multi-Modal Network:** ResNet-18-PT visual backbone fused with a tabular 2-feature Multi-Layer Perceptron (MLP). Initialized via MedicalNet pretraining. Inputs: 3D MRI Volume + Demographics (Z-scored Age and Sex).
- **Random Forest (Baseline):** scikit-learn RandomForestClassifier (300 trees, $\text{max_depth} = 6$, $\text{min_samples_leaf} = 5$, $\text{class_weight} = \text{balanced}$). Inputs: Structural Tabular Features (Age, Sex, eTIV, nWBV, ASF).

Training Dynamics and Optimization

Deep CNN models trained from scratch were optimized over a maximum bound of 60 epochs (minimum execution floor of 35 epochs), implementing an early stopping warm-up ban across the first 20 epochs with an active patience window of 15 epochs. The optimization framework utilized the Adam optimizer paired with a weight decay constraint of $1e-4$ and a label smoothing regularization factor of 0.1. The learning rate followed a Cosine Annealing trajectory down to a minimum floor of $1e-6$, starting from an initial learning rate of $1e-3$ for ResNet variants and $5e-4$ for DenseNet structures. The effective batch size was stabilized at 8 using gradient accumulation steps.

For the transfer learning pipelines (ResNet-18-PT and Multi-Modal), a two-phase fine-tuning schedule was developed:

- Phase 1: The pretrained feature extractor was frozen for 10 epochs while the classification head was optimized at a learning rate of $1e-3$.
- Phase 2: The entire network was unfrozen across 40 epochs, enforcing localized discriminative learning rates across different architectural blocks (layer3: $1e-5$, layer4: $5e-5$, head: $1e-4$) with a patience of 12 epochs.

The definitive model checkpoint selection criterion rejected raw optimization metrics like $\min(\text{val_loss})$ or $\max(\text{val_auc})$. Instead, it utilized a robust composite fitness score:

$$\text{Score} = \text{val_auc} - (0.3 * \text{val_loss})$$

This mathematical regularizer penalizes overconfident incorrect predictions, stabilizing model selection over constrained medical datasets.

8. Results

To evaluate the models under diverse clinical requirements, a comprehensive threshold sweep (including Youden-optimal points) was executed across all architectures using pooled out-of-fold (OOF) predictions. This analysis revealed that transfer learning (ResNet-18-PT) and Multi-Modal networks consistently outperformed models trained from scratch. Furthermore, the sweep

confirmed that the default threshold ($t=0.50$) naturally provides the optimal clinical balance between sensitivity and specificity for these top-performing models.

Performance Profile at Optimal Classification Threshold ($t = 0.50$)

The table below displays the cross-model performance metrics at this selected baseline threshold, which yielded the best diagnostic trade-off for the leading candidates:

Model Pipeline	OOF AUC	Accuracy	Sensitivity	Specificity	F1-Score
ResNet-18 (Scratch)	0.681	0.592	0.667	0.524	0.590
DenseNet-121 (Scratch)	0.663	0.628	0.581	0.670	0.627
ResNet-18-PT	0.674	0.628	0.753	0.515	0.623
Multi-Modal	0.672	0.633	0.731	0.544	0.630
Random Forest (Fair)	0.704	0.628	0.634	0.621	0.628

Performance Profile at Youden-Optimal Threshold (Per-Fold Mean plus/minus SD)

The table below details the cross-validated metrics evaluated at each architecture's customized optimal operating point:

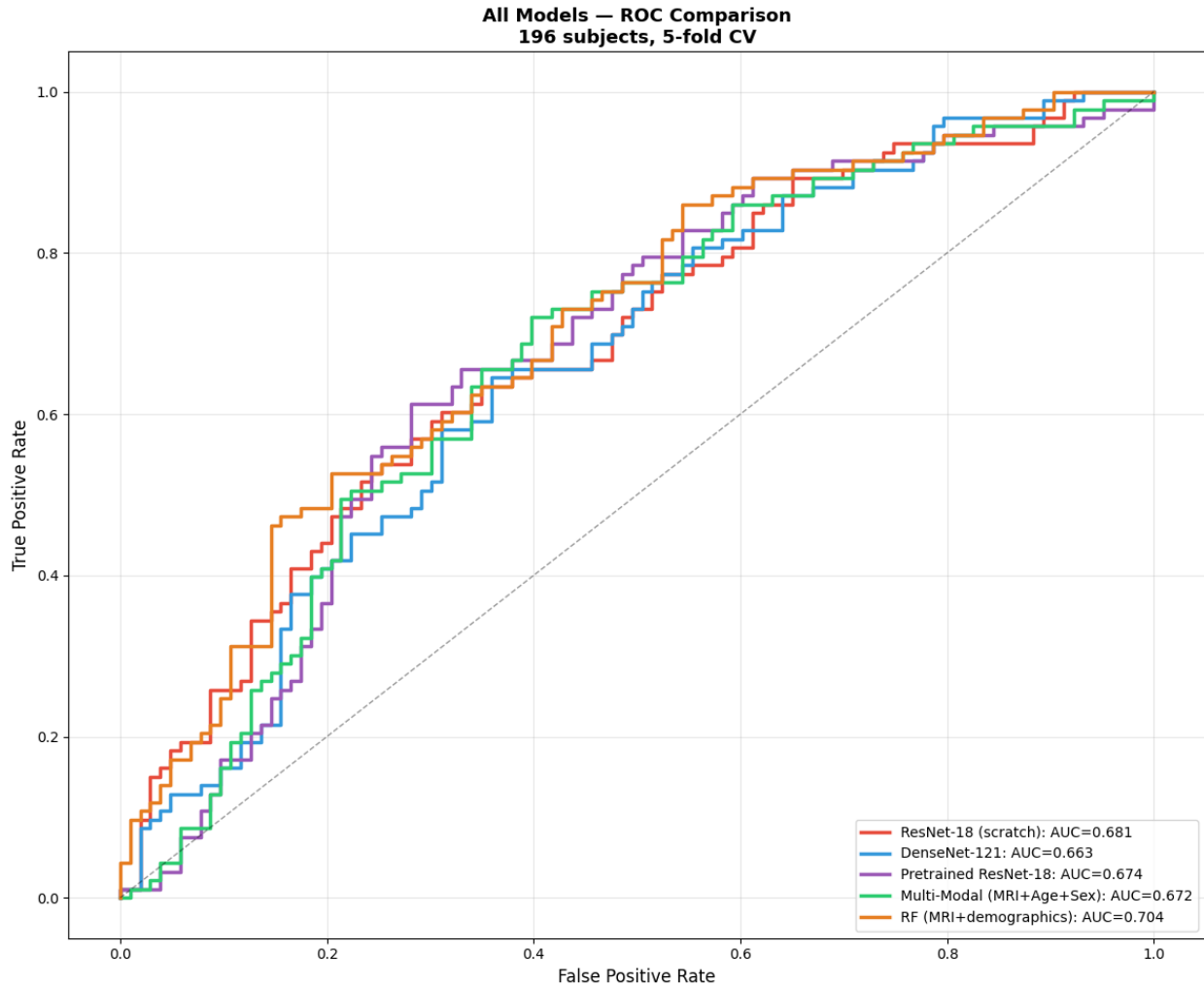
Model Pipeline	Youden t	Mean plus/minus SD AUC	Accuracy	Sensitivity	Specificity	F1-Score
ResNet-18 (Scratch)	0.553	0.721 plus/minus 0.036	0.648	0.598	0.689	0.633
DenseNet-121	0.487	0.666 plus/minus 0.063	0.644	0.646	0.643	0.642

ResNet-18-PT	0.587	0.688 plus/minus 0.091	0.669	0.613	0.719	0.665
Multi-Modal	0.522	0.684 plus/minus 0.105	0.659	0.722	0.601	0.650
Random Forest (Fair)	0.614	0.710 plus/minus 0.121	0.669	0.526	0.797	0.661

Granular Decision Threshold Sweep for the Premier Model (Pooled OOF)

Although threshold sweeps were computed for all models, the table below details the diagnostic trade-offs specifically for the best-performing architecture (ResNet-18-PT). This breakdown illustrates why the default threshold ($t=0.50$) was selected as its ideal clinical operating point:

Operating Threshold (t)	Pooled OOF Accuracy	Sensitivity	Specificity	F1-Score
0.35	0.597	0.914	0.311	0.559
0.40	0.622	0.892	0.379	0.598
0.45	0.617	0.828	0.427	0.603
0.50 (Selected)	0.628	0.753	0.515	0.623
0.55	0.628	0.667	0.592	0.627
0.60	0.648	0.559	0.728	0.645



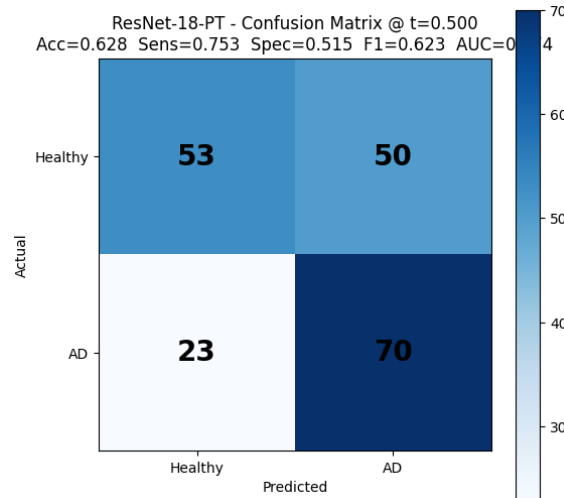
Definitive Best Model Selection Analysis

Although the Multi-Modal and ResNet-18-PT pipelines achieved closely matched discrimination at $t=0.50$ (AUC = 0.672 vs 0.674), ResNet-18-PT was selected as the definitive clinical model based on two key advantages:

- **Cross-Validated Stability:** It demonstrated superior generalization, yielding a lower per-fold standard deviation (± 0.091 vs ± 0.105) compared to the fused network.
- **Architectural Parsimony:** By relying solely on neuroimaging, it eliminates dependency on tabular demographic data at deployment without sacrificing performance.

Clinically, this selected profile maximizes screening utility (Sensitivity = 0.753), successfully capturing 70 out of 93 true AD cases within the cohort. The final pooled out-of-fold confusion matrix at $t=0.50$ is detailed below:

- Actual Healthy (n=103): 53 predicted as Healthy, 50 predicted as AD.
- Actual Alzheimer (n=93): 23 predicted as Healthy, 70 predicted as AD.



9. Discussion

The primary empirical finding of this study is that across all pipelines, performance converged into a narrow statistical band (AUC 0.66–0.72). This convergence indicates a small-data performance plateau; algorithmic capacity is not the bottleneck. Rather, a single cross-sectional 3D MRI per subject (n=196) lacks the statistical variation required to establish a highly separable decision boundary.

Within this plateau, domain-specific transfer learning (ResNet-18-PT) proved invaluable for cross-validated stability rather than absolute peak metrics. While the scratch-trained model exhibited severe variance—achieving an artificially high AUC (0.766) in Fold 5 but collapsing in Fold 2 (Sensitivity = 0.167)—the pretrained network remained robust across all splits. In constrained medical datasets, this cross-validated consistency is a far more reliable indicator of clinical utility than highly volatile point-estimates.

Feature Subset Diagnostic Analysis

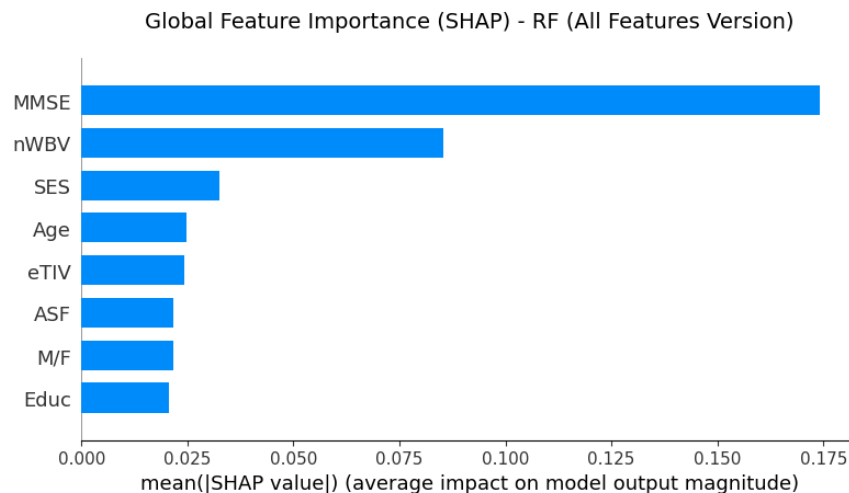
To understand the performance dynamics of the Random Forest baseline, a localized feature-subset diagnostic experiment was executed at $t = 0.50$:

Feature Configuration	Split	Contained Metrics	OOF AUC	Accuracy	Sensitivity
Imaging-Derived Only		eTIV, nWBV, ASF	0.676	0.612	0.634
Imaging + Demographics (Fair)		Age, Sex, eTIV, nWBV, ASF	0.704	0.628	0.634

All Features (Unfair Proxy)	Tabular + Demographics + MMSE	0.884	0.811	0.785
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To understand the baseline dynamics, feature subsets were analyzed. Notably, the "fair" Random Forest (using only volumetric and demographic features) achieved an AUC of 0.704, tightly overlapping with the CNNs. This reinforces that the small-data plateau affects classical and deep learning paradigms equally; CNNs could not extract significantly better separable features from the raw volumes than classical FreeSurfer morphometry.

Conversely, including the MMSE score caused an artificial jump in performance (AUC = 0.884). As visually confirmed by the Random Forest feature importance analysis, the MMSE feature overwhelmingly dominates the decision tree splits, dwarfing all structural MRI metrics. Because MMSE directly measures cognitive decline, it acts as a proxy for the target labels, transforming the task from "imaging detection" to "predicting a cognitive score."



Similarly, multi-modal CNN fusion provided no measurable benefit over the pure-imaging network, as demographics act as weak independent predictors in an already age-restricted cohort (≥ 58 years). This completely justifies selecting the parsimonious imaging-only architecture.

Pipeline Limitations

Several notable limitations apply to this study:

1. **Sample Constraints:** The sample size ($n = 196$) remains the primary limiter, which cannot be fully resolved by architectural regularizers or transfer learning mechanics.
2. **Acquisition Biases:** The presence of demographic sex imbalances (130 females vs. 66 males) combined with single-site scanner protocol uniformities introduces potential acquisition biases.
3. **Target Simplification:** The target label setup simplifies the problem into a broad binary task ($\text{CDR} = 0$ vs. $\text{CDR} > 0$), bypassing more complex clinical scenarios such as Mild Cognitive Impairment (MCI) sub-classification or longitudinal progression mapping.

10. Conclusions and Future Work

Through a rigorously controlled 3D volumetric cross-validation framework, this study concludes that a ResNet-18 architecture initialized with domain-specific MedicalNet weights offers the most stable and reliable pipeline for Alzheimer's detection under constrained sample sizes.

Future Work

To extend and enhance the clinical utility of this framework, future iterations will focus on four target areas:

- **Dataset Expansion:** Incorporating large datasets (e.g., ADNI, OASIS-3).
- **Targeted ROI Cropping:** Replacing the global 128x128x128 isotropic downsampling with target bounding crops focused on localized anatomical structures with high AD effect sizes, such as the hippocampus and medial temporal lobe structures.
- **Longitudinal MCI Tracking:** Transitioning from broad binary detection to granular multi-class prediction (including Mild Cognitive Impairment, CDR=0.5) to map the longitudinal progression of the disease over time.
- **Enhanced Multi-Modal Fusion:** As revealed by the post-hoc SHAP analysis of the classical models, Socioeconomic Status (SES) carries unexpectedly high predictive weight. Future deep learning architectures will aim to incorporate SES alongside Age and Sex. To prevent these low-dimensional tabular features from being overshadowed by the high-dimensional 512-vector imaging embeddings (Modality Dominance), advanced fusion mechanisms such as cross-attention layers will be explored instead of simple late-stage concatenation.

11. Ethics and Data Privacy

The dataset utilized is from the open-access OASIS-1 repository, accessed under the official Data Use Agreement permitting non-commercial scientific analysis. The curated data is completely de-identified, contains no Protected Health Information (PHI), and all MRI volumes have undergone facial defacing to ensure total subject anonymity. The transfer learning weights were sourced from the open MedicalNet repository under a research-only license.

12. Code and Reproducibility

- **Core Execution:** Alzheimer_KFold_Ensemble_v3_1.ipynb (Executed via Google Colab).
- **Hardware Profiling:** Google Colab, NVIDIA L4 Tensor Core GPU, CUDA compilation version 12.8.
- **Software Dependencies & Version Controls:** Python 3.11, torch 2.11.0 + cu128, monai 1.5.2, scikit-learn, pandas, numpy, matplotlib, shap, nibabel, and huggingface_hub (for MedicalNet weight synchronization).
- **Deterministic Guardrails:** System-wide deterministic execution was guaranteed by pinning a constant seed (seed = 42) across NumPy, PyTorch, and MONAI, while forcing deterministic cuDNN backends (torch.use_deterministic_algorithms(True)). Automatic mixed precision (AMP) was disabled to eliminate floating-point non-determinism. The multi-seed ensembling initializations (seeds 42, 7, and 21) modified model parameter state generation without altering data cross-validation layouts.

- **Data Routing & Execution Notes:** To reproduce the results, the OASIS-1 demographic CSV must be placed in the project root directory. Next, an auxiliary data ingestion script must be executed; this script parses the nested OASIS-1 release structures, extracts the raw FreeSurfer .mgz volumes from individual folders, and flattens them into a centralized working directory (alzheimers_project/). The notebook then automatically maps the diagnostic labels from the CSV to this flat directory and exports out-of-fold inference arrays as compressed NumPy structures (*.npz).
- **AI Assistance Disclosure:** Large Language Models (specifically Anthropic's Claude and Google's Gemini) were utilized as pair-programming assistants during the development phase. The LLM was employed strictly to optimize code structure, accelerate debugging, and refine the grammatical flow of this report, while all core architectural decisions and experimental designs remain the original work of the author.

13. References

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